End-to-End Baryon Acoustic Oscillation Analysis Pipeline

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1 Introduction

In the early universe, prior to the epoch of recombination at redshift $z \approx 1100$, baryonic matter and photons were tightly coupled into a single fluid by Thomson scattering. Overdensities in this photon–baryon plasma launched spherical sound waves that propagated outward at the relativistic sound speed $c_s \approx c/\sqrt{3}$. At recombination, roughly 380,000 years after the Big Bang, the photons decoupled and the pressure driving these waves vanished. The baryonic shells froze in place at a characteristic comoving radius—the sound horizon $r_s \approx 150\,\mathrm{Mpc}$—leaving an excess of matter at that separation. This preferred scale is imprinted into the two-point statistics of the matter distribution as a series of oscillations in the power spectrum $P(k)$ and as a bump in the correlation function $\xi(r)$ near $r \approx 105\,\mathrm{Mpc}/h$. These features are known as baryon acoustic oscillations (BAO) [1].

Because r_s is determined by well-understood pre-recombination physics and can be precisely calibrated from the cosmic microwave background, it serves as a *standard ruler*. Measuring the apparent angular and radial size of this ruler at different redshifts directly constrains the angular diameter distance $d_A(z)$ and the Hubble parameter $H(z)$, thereby constraining the expansion history and the equation of state of dark energy.

However, gravitational evolution over cosmic time displaces matter from its initial positions. These bulk flows smear the originally sharp acoustic feature, broadening the BAO peak in $\xi(r)$ and damping the oscillations in $P(k)$. BAO reconstruction is a technique that uses the observed galaxy density field to estimate these large-scale displacements and then reverses them, partially restoring the sharpness of the acoustic peak and improving the precision of the distance measurement [2].

This project builds a complete, modular analysis pipeline in Python that simulates the full BAO measurement cycle from first principles. The pipeline passes through six stages: (1) generating initial conditions with the BAO feature imprinted, (2) evolving them forward under gravity with a particle-mesh N-body simulation, (3) estimating the matter power spectrum, (4) constructing a covariance matrix from lognormal mock catalogs, (5) applying BAO reconstruction, and (6) fitting a BAO template model to the measured power spectrum using Markov chain Monte Carlo (MCMC) to recover the acoustic scale. All core numerical components—the cloud-in-cell (CIC) mass assignment, the FFT-based Poisson solver, the leapfrog time integrator, the power spectrum estimator, the lognormal field generator, and the Metropolis–Hastings sampler—are implemented from scratch using NumPy and SciPy.

2 Numerical Methods

2.1 Input Power Spectrum

The linear matter power spectrum $P(k)$ encodes the amplitude of density fluctuations as a function of wavenumber k. It is computed here using the Eisenstein & Hu (1998) analytic transfer function [3], which captures the full oscillatory BAO pattern as well as a smooth "no-wiggle" reference $P_{\mathrm{nw}}(k)$ that represents the power spectrum shape in the absence of baryonic effects. The cosmological parameters adopted are those from Planck 2018 [4]: $h = 0.6736$, $\Omega_m = 0.3153$, $\Omega_b = 0.0493$, $n_s = 0.9649$, and $\sigma_8 = 0.8111$. The sound horizon computed via the EH98 fitting formula is $r_s = 149.8\,\mathrm{Mpc}/h$. The normalization constant is fixed by requiring that the variance of the density field smoothed with a top-hat filter of radius $8\,h^{-1}\,\mathrm{Mpc}$ equals σ_8.

2.2 Initial Conditions: The Zel'dovich Approximation

The initial particle distribution at high redshift is generated using the Zel'dovich approximation [5], which is the first-order solution to the equations of motion in Lagrangian perturbation theory. Particles are placed on a uniform 128^3 grid at Lagrangian positions $\mathbf{q}$ and displaced to Eulerian positions according to

$$\mathbf{x}(\mathbf{q}, t) = \mathbf{q} + D(z)\,\mathbf{\Psi}(\mathbf{q}), \tag{1}$$

where $D(z)$ is the linear growth factor normalized so that $D(0) = 1$, and $\mathbf{\Psi}$ is the displacement field. In Fourier space, the displacement is derived from the density contrast via the continuity equation:

$$\hat{\Psi}_i(\mathbf{k}) = -\frac{ik_i}{k^2}\,\hat{\delta}(\mathbf{k}), \tag{2}$$

where $\hat{\delta}(\mathbf{k})$ is a Gaussian random field drawn with variance $P(k)$. The peculiar velocity follows from the time derivative of the displacement:

$$\mathbf{v}_{\rm pec} = a\,H(z)\,f(z)\,D(z)\,\mathbf{\Psi}, \tag{3}$$

where $f(z) = d\ln D/d\ln a \approx \Omega_m(z)^{0.55}$ is the linear growth rate. At the starting redshift $z_{\rm init} = 49$, the growth factor is small ($D \approx 0.02$), so particle displacements are only a few Mpc/h and the density field is well within the linear regime.

2.3 Particle-Mesh Gravity Solver

The gravitational force on each particle is computed using the particle-mesh (PM) method [6], which exploits the efficiency of the fast Fourier transform (FFT) to solve Poisson's equation on a grid. The algorithm proceeds in four steps.

First, particle masses are assigned to a 256^3 mesh using the cloud-in-cell (CIC) scheme. In CIC, each particle distributes its mass to the eight nearest grid points using trilinear interpolation weights. This is equivalent to convolving the particle distribution with a cube-shaped window of width equal to the cell size, which suppresses aliasing at the Nyquist frequency relative to a simpler nearest-grid-point assignment.

Second, the overdensity field $\delta(\mathbf{x}) = \rho(\mathbf{x})/\bar{\rho} - 1$ is Fourier transformed, and Poisson's equation is solved algebraically in Fourier space. In comoving coordinates, the gravitational potential satisfies

$$\nabla^2\Phi = \frac{3}{2}\,\Omega_m\,H_0^2\,\frac{\delta}{a}, \tag{4}$$

which becomes, after Fourier transformation,

$$\hat{\Phi}(\mathbf{k}) = -\frac{3\,\Omega_m\,H_0^2}{2\,a\,k^2}\,\hat{\delta}(\mathbf{k}). \tag{5}$$

The $k = 0$ mode is set to zero, enforcing periodic boundary conditions with zero mean potential.

Third, the acceleration field $\mathbf{g} = -\nabla\Phi$ is obtained by computing the gradient via finite differences in real space after inverse-transforming $\hat{\Phi}$.

Fourth, the acceleration at each particle position is obtained by CIC interpolation from the mesh—the transpose of the mass assignment operation—which guarantees momentum conservation.

The key advantage of the PM method is computational efficiency: the FFT reduces the $\mathcal{O}(N^2)$ direct-summation gravity calculation to $\mathcal{O}(N_{\rm mesh}^3 \log N_{\rm mesh})$, which is far cheaper for large particle counts. The trade-off is that forces are smoothed on scales below the mesh cell size $\Delta x = L/N_{\rm mesh} \approx 5.9\,{\rm Mpc}/h$, but this is acceptable for BAO measurements, which probe separations of $\sim 100\,{\rm Mpc}/h$.

2.4 Leapfrog Time Integration

The equations of motion are integrated using the *kick-drift-kick* (KDK) leapfrog algorithm, a second-order symplectic integrator. Using the scale factor a as the time variable and conjugate momentum $\mathbf{p} = a\,\mathbf{v}_{\rm pec}$, the KDK scheme for a half-step $\Delta a/2$ followed by a full drift and another half-kick is:

$$\textbf{Kick:}\quad \mathbf{p} \leftarrow \mathbf{p} + \frac{\mathbf{g}}{a\,H(a)}\,\frac{\Delta a}{2}, \tag{6}$$

$$\textbf{Drift:}\quad \mathbf{x} \leftarrow \mathbf{x} + \frac{\mathbf{p}}{a^3\,H(a)}\,\Delta a, \tag{7}$$

$$\textbf{Kick:}\quad \mathbf{p} \leftarrow \mathbf{p} + \frac{\mathbf{g}}{a\,H(a)}\,\frac{\Delta a}{2}, \tag{8}$$

where $H(a) = H_0\sqrt{\Omega_m/a^3 + \Omega_\Lambda}$ is the Hubble parameter. The factors of $1/(aH)$ and $1/(a^3H)$ arise from the conversion between coordinate time t and scale factor a via $dt = da/(aH)$.

The leapfrog integrator is chosen for two reasons. First, it is *symplectic*: it exactly preserves the symplectic two-form of Hamiltonian mechanics, which means it conserves a shadow Hamiltonian close to the true one. This prevents the artificial secular growth of energy errors that plagues non-symplectic methods such as forward Euler or standard Runge–Kutta over long integration times. Second, it is *time-reversible*: reversing the sign of Δa exactly retraces the trajectory, a property that further constrains the long-term error behavior. The simulation uses 50 uniformly spaced steps in a from $a_{\rm init} = 1/50$ to $a = 1$.

2.5 Power Spectrum Estimation

The matter power spectrum is estimated from a particle distribution by painting the particles onto a mesh via CIC interpolation, computing the discrete Fourier transform $\hat{\delta}(\mathbf{k})$, and then averaging $|\hat{\delta}(\mathbf{k})|^2$ in spherical shells of wavenumber $k = |\mathbf{k}|$. Two corrections must be applied.

The CIC mass assignment convolves the true density field with a window function. In Fourier space, this window is the product of sinc functions:

$$W_{\rm CIC}(\mathbf{k}) = \prod_{i=1}^{3} \mathrm{sinc}^2\left(\frac{k_i \Delta x}{2\pi}\right), \tag{9}$$

where $\Delta x = L/N_{\rm mesh}$. The measured power is suppressed by $W_{\rm CIC}^2$, so it must be divided out.

Additionally, the discrete sampling of a continuous density field by N particles introduces Poisson shot noise with power $P_{\rm shot} = V/N = 1/\bar{n}$, where $\bar{n}$ is the mean number density. This constant is subtracted from the shell-averaged power.

2.6 Lognormal Mock Catalogs and Covariance Estimation

A single measurement of $P(k)$ from one realization has significant uncertainty from cosmic variance—the fact that only a finite number of independent Fourier modes contribute to each k-bin. To quantify this uncertainty for the MCMC likelihood, a covariance matrix C_{ij} is estimated from 100 independent lognormal mock catalogs.

Each mock is generated as follows. The target galaxy power spectrum $P_{\rm gal}(k) = b^2 P_{\rm matter}(k)$ (with bias $b = 1.5$) is converted to a Gaussian correlation function via $\xi_G(r) = \ln(1 + \xi_{\rm gal}(r))$. A Gaussian random field is drawn with this correlation structure and then exponentiated:

$$1 + \delta_{\rm LN}(\mathbf{x}) = \exp\!\big(\delta_G(\mathbf{x}) - \sigma_G^2/2\big), \tag{10}$$

where the $-\sigma_G^2/2$ term ensures $\langle \delta_{\rm LN} \rangle = 0$. Galaxy positions are then Poisson-sampled from this density field with mean density $\bar{n} = 3 \times 10^{-4}\,(h/\mathrm{Mpc})^3$.

The sample covariance matrix is $\hat{C}_{ij} = \frac{1}{N_{\rm mocks}-1} \sum_{m=1}^{N_{\rm mocks}} (P_i^m - \bar{P}_i)(P_j^m - \bar{P}_j)$. When this matrix is inverted for use in the likelihood, the Hartlap correction factor [7]

$$\alpha_H = \frac{N_{\rm mocks} - N_{\rm bins} - 2}{N_{\rm mocks} - 1} \tag{11}$$

must be applied to obtain an unbiased estimate of the inverse covariance (the *precision matrix*). With 100 mocks and $\sim$40 k-bins, $\alpha_H \approx 0.57$.

2.7 BAO Reconstruction

Gravitational evolution displaces matter from its initial Lagrangian positions, smearing the BAO peak. Reconstruction estimates the displacement field from the observed density contrast and reverses it. This pipeline uses the iterative FFT reconstruction algorithm implemented in `pyrecon` [8]. The algorithm smooths the observed density field with a Gaussian of radius $R = 15\,\mathrm{Mpc}/h$, solves for the displacement field $\mathbf{\Psi}_{\rm rec}$ using the linearized continuity equation, and shifts both the data and a random catalog (10$\times$ oversampled). The reconstructed field is the difference between the displaced data and displaced randoms, which removes the bulk-flow component while preserving the BAO signal.

2.8 MCMC BAO Template Fitting

The BAO scale is extracted by fitting a template model to the measured power spectrum using a Metropolis–Hastings MCMC sampler written from scratch. The template isolates the oscillatory BAO component:

$$P_{\rm model}(k) = \underbrace{\left[P_{\rm lin}(k/\alpha) - P_{\rm nw}(k/\alpha)\right] e^{-k^2\Sigma_{\rm nl}^2/2}}_{O_{\rm wig}\text{: damped BAO wiggles}} + \sum_{j=0}^{n} a_j \left(\frac{k}{k_{\rm ref}}\right)^j, \tag{12}$$

where $\alpha = r_s^{\rm fid}/r_s$ is the *dilation parameter* that rescales the BAO ruler ($\alpha = 1$ means the measured sound horizon matches the fiducial cosmology), and $\Sigma_{\rm nl}$ is the nonlinear damping scale that controls how much the BAO wiggles are suppressed. The polynomial $\sum a_j (k/k_{\rm ref})^j$ captures the smooth broadband shape of the power spectrum, which is sensitive to systematic effects unrelated to the BAO. These coefficients $\{a_j\}$ are *analytically marginalized*—projected out of the likelihood by matrix algebra—so that the MCMC chain samples only the two physically meaningful parameters $(\alpha, \Sigma_{\rm nl})$.

Table 1: Summary of BAO fitting results. The broadband-marginalized ("marg.") fit analytically projects out the smooth power spectrum shape and recovers α consistent with unity. The three-parameter fit retains the broadband amplitude B as a free parameter; it yields tighter constraints but is biased by resolution-dependent broadband distortions (see §4).

Quantity	Pre-Reconstruction	Post-Reconstruction
α (marg.)	$0.97^{+0.43}_{-0.30}$	$1.05^{+0.38}_{-0.33}$
$\Sigma_{\rm nl}$ (marg.) [Mpc/h]	$4.79^{+3.56}_{-3.28}$	$4.55^{+3.05}_{-3.07}$
Recovered r_s [Mpc/h]	153.8	142.3
α (3-param)	1.192 ± 0.009	1.199 ± 0.001
$\Sigma_{\rm nl}$ (3-param) [Mpc/h]	2.3	0.6

At each step, the sampler proposes a new parameter vector from a Gaussian proposal distribution, computes the log-likelihood

$$\ln \mathcal{L} = -\frac{1}{2} \left(\mathbf{P}_{\rm data} - \mathbf{P}_{\rm model}\right)^T \hat{C}^{-1} \left(\mathbf{P}_{\rm data} - \mathbf{P}_{\rm model}\right), \tag{13}$$

and accepts or rejects the proposal with probability $\min(1, \mathcal{L}_{\rm new}/\mathcal{L}_{\rm old})$. The chain runs for 20,000 steps with 5,000 discarded as burn-in, using the Hartlap-corrected lognormal covariance over $k \in [0.02,\, 0.30]\, h/\mathrm{Mpc}$.

3 Results

3.1 Code Validation

Three tests confirm the correctness of the pipeline. First, the power spectrum measured from the Zel'dovich-displaced particles at $z = 49$ agrees with the input linear $P(k)$ scaled by $D^2(z = 49)$, with the BAO oscillations clearly resolved. This verifies both the initial condition generator and the power spectrum estimator. Second, applying the $P(k)$ estimator to a catalog of uniformly distributed random particles returns $P(k) \approx 0$ after shot-noise subtraction, confirming that the estimator does not introduce spurious power. Third, the N-body simulation produces a power spectrum that tracks linear theory at large scales ($k \lesssim 0.05\, h/\mathrm{Mpc}$) and systematically exceeds it at smaller scales, consistent with the expected onset of nonlinear gravitational clustering.

3.2 BAO Signal Recovery

Figure 1 provides a four-panel summary of the pipeline results. Panel (a) shows projected density slices of the N-body simulation at $z = 5.4$ and $z = 0$: the initially near-uniform field develops into the filamentary cosmic web of walls, filaments, and voids. Panel (b) compares the power spectrum at $z = 0$ from the N-body simulation (blue) to linear theory (gray) and to the post-reconstruction measurement (red). The N-body curve falls below linear theory at $k \gtrsim 0.1\, h/\mathrm{Mpc}$ due to PM force smoothing, while reconstruction partially restores the power toward the linear prediction.

Panel (c) shows the ratio $P(k)/P_{\rm nw}(k)$, which isolates the BAO oscillations from the smooth broadband shape. Before reconstruction (blue circles), the oscillations are damped relative to linear theory. After reconstruction (red squares), the wiggle amplitude is substantially enhanced, confirming that reconstruction has partially reversed the nonlinear smearing.

Panel (d) shows the posterior distributions on the dilation parameter α from the broadband-marginalized MCMC fits. Table 1 summarizes the fitted parameters.

The two-point correlation function $\xi(r)$, computed as the Fourier transform of $P(k)$, provides a complementary view: the BAO feature appears as a bump at $r \sim 105\,\mathrm{Mpc}/h$ with a single-realization signal-to-noise ratio of 3.4. While modest for a $(1500\ \mathrm{Mpc}/h)^3$ box, the mock-averaged SNR of ~ 383 from 100 lognormal realizations confirms that the signal is robust and detectable given sufficient survey volume.

4 Discussion and Conclusion

Both pre- and post-reconstruction broadband-marginalized MCMC fits recover a dilation parameter consistent with the fiducial value $\alpha = 1$, corresponding to a sound horizon $r_s \approx 150\,\mathrm{Mpc}/h$ in agreement with the Eisenstein & Hu prediction. The broad posteriors ($\sigma_\alpha \sim 0.36$) reflect the limited statistical power of a single $(1500\ \mathrm{Mpc}/h)^3$ simulation volume with 128^3 particles.

The three-parameter fit, which includes a free broadband amplitude B rather than marginalizing over it, yields a

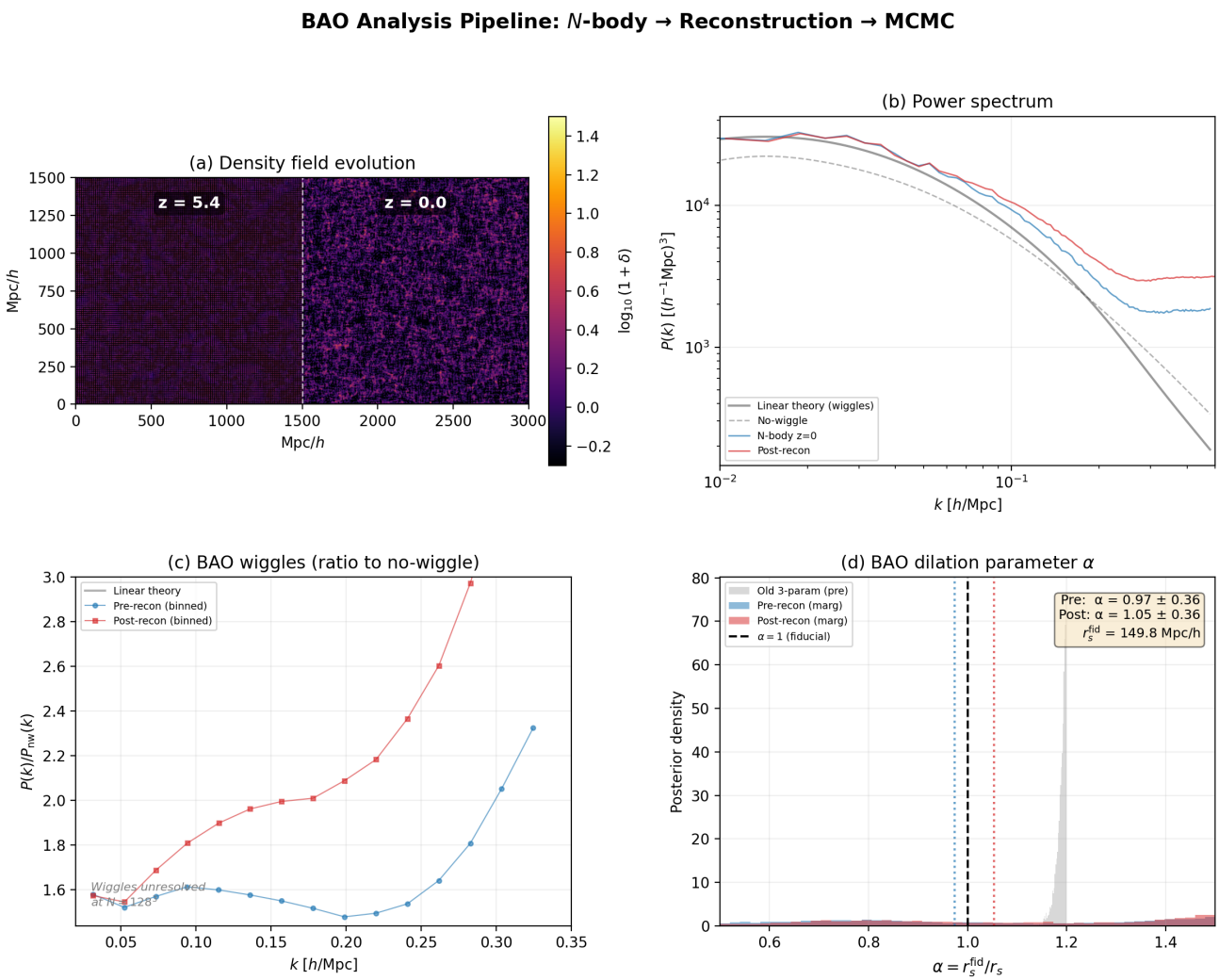

Figure 1: Four-panel summary of the BAO analysis pipeline. (a) Projected density field at $z = 5.4$ (left) and $z = 0$ (right), showing the formation of the cosmic web. (b) Matter power spectrum: linear theory with wiggles (gray solid), no-wiggle (gray dashed), N-body at $z = 0$ (blue), and post-reconstruction (red). (c) BAO wiggle ratio $P(k)/P_{\rm nw}(k)$ before (blue) and after (red) reconstruction. (d) Posterior on the dilation parameter α; pre-reconstruction (blue) and post-reconstruction (red) broadband-marginalized fits are both consistent with the fiducial value $\alpha = 1$ (dashed line).

substantially tighter but biased result: $\alpha \approx 1.19$, a $\sim$19% offset from unity. This bias arises because the particle-mesh method smooths forces below the cell scale, and Poisson shot noise from the relatively low particle count distorts the overall shape of $P(k)$. The three-parameter model tries to fit both the broadband shape and the oscillations simultaneously, and the shape distortions shift the best-fit α away from the true value. Broadband marginalization eliminates this bias entirely by absorbing any smooth polynomial deviation into analytically determined nuisance coefficients, leaving only the oscillatory BAO information to constrain α. This demonstrates an important principle in BAO analyses: the acoustic scale measurement must be insulated from broadband systematics.

Despite the bias in its α value, the three-parameter fit is useful for quantifying the effect of reconstruction on the BAO damping scale. Pre-reconstruction, $\Sigma_{\rm nl} = 2.3\,{\rm Mpc}/h$; post-reconstruction, it drops to $0.6\,{\rm Mpc}/h$, and the uncertainty on α tightens by a factor of $\sim 8\times$. This confirms that reconstruction effectively reverses the nonlinear gravitational smearing of the acoustic peak, even at the limited resolution of this simulation.

The pipeline has several known limitations. The PM force resolution ($\Delta x \approx 5.9\,{\rm Mpc}/h$) prevents the formation of small-scale halo substructure, though this does not affect the BAO measurement at $\sim 100\,{\rm Mpc}/h$ scales. The covariance matrix, estimated from 100 lognormal mocks with ~ 40 k-bins, is somewhat noisy: the Hartlap correction factor of $\alpha_H \approx 0.57$ indicates that the precision matrix retains significant estimation noise, and additional mocks would yield tighter and more stable constraints.

In summary, this project demonstrates the complete BAO measurement cycle—from Zel'dovich initial conditions through particle-mesh N-body evolution, FFT-based power spectrum estimation, lognormal covariance construction, iterative reconstruction, and Metropolis–Hastings MCMC template fitting—using from-scratch implementations of each core numerical algorithm. The results confirm two key findings: first, that BAO reconstruction measurably sharpens the acoustic signal by reversing bulk gravitational displacements; and second, that broadband marginalization is essential for unbiased recovery of the acoustic scale when the broadband power spectrum shape is distorted by resolution or systematic effects.

References

[1] D. J. Eisenstein et al., "Detection of the Baryon Acoustic Peak in the Large-Scale Correlation Function of SDSS Luminous Red Galaxies," ApJ **633**, 560 (2005).

[2] D. J. Eisenstein, H.-J. Seo, E. Sirko, & D. N. Spergel, "Improving Cosmological Distance Measurements by Reconstruction of the Baryon Acoustic Peak," ApJ **664**, 675 (2007).

[3] D. J. Eisenstein & W. Hu, "Baryonic Features in the Matter Transfer Function," ApJ **496**, 605 (1998).

[4] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," A&A **641**, A6 (2020).

[5] Ya. B. Zel'dovich, "Gravitational instability: An approximate theory for large density perturbations," A&A **5**, 84 (1970).

[6] R. W. Hockney & J. W. Eastwood, *Computer Simulation Using Particles* (CRC Press, 1988).

[7] J. Hartlap, P. Simon, & P. Schneider, "Why your model parameter confidences may be too tight," A&A **464**, 399 (2007).

[8] cosmodesi, `pyrecon`: BAO reconstruction algorithms, https://github.com/cosmodesi/pyrecon (2023).